

Junxiu TANG

CONTACT INFORMATION

Center for Sci of Sci & Innovation (CSSI)
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Evanston, Illinois, U.S., 60201

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Updated: May, 2026

RESEARCH INTERESTS

My research explores the transformation of data into **valuable insights**, **compelling narratives**, and **aesthetic expressions**. It encompasses advanced data visualization techniques, rigorous scientific logic, and artistic presentation methods. Specifically, my past research lies in the design and interaction of data-driven communication, storytelling, and event sequence visualization. I am also keen to explore relationship between data and art.

Keywords: Data Visualization, Visual Analytics, Fixed-order Event Sequence, Embedded Visualization, Data-driven Art, Data-driven Communication, Data Video, Storytelling, Human-AI Collaboration

WORK

Northwestern University, Evanston, US 06.2025–Present
Center for Science of Science & Innovation (CSSI)
Kellogg School of Management
Post-Doctoral Researcher

Zhejiang University, Hangzhou, China 01.2025–05.2025
Interactive Data Group
College of Computer Science and Technology
Research Assistant

EDUCATION

Zhejiang University, Hangzhou, China 09.2019–12.2024
Department of Computer Science and Technology
College of Computer Science and Technology
Ph.D., with Prof. Yingcai Wu, at ZJUIDG
Thesis: Theoretical and Applied Research of Visualizations for Fixed-order Event Sequence

Inria, Palaiseau, France 11.2023–06.2024
Inria Saclay Center
Visiting Ph.D. Student, with Dr. Petra Isenberg, at Aviz
Research Focus: Embedded Visualizations in Swimming Videos

Zhejiang University, Hangzhou, China 09.2014–06.2019
Department of Digital Media Technology
College of Computer Science and Technology
B.Eng.
Research Focus: Design Guidelines on Augmenting Short-form Data Videos

RESEARCH INTERSHIPS

JD Intelligent Cities Research, Jing Dong Technology 02.2022–06.2022
Beijing, China
Research Intern, Software Engineer
Building visual analytics systems for urban computing, especially intelligent transportation and disease control. Supervised by Dr. Jie Bao and Prof. Yu Zheng.

**IDEA Lab, Alibaba-Zhejiang University Joint Research
Institute of Frontier Technology (AZFT)**

Hangzhou, China

09.2019–12.2021

Research Intern

Cooperated with DAMO Academy and Cainiao

Cainiao Smart Logistics, Alibaba Group

Hangzhou, China

09.2020-12.2021

Research Intern, UI&UX Designer

Building the visual analytics system for intelligent logistics,
especially on order processing monitoring and optimization.

**Alibaba DAMO Academy for Discovery, Adventure,
Momentum and Outlook, Alibaba Group**

Hangzhou, China

11.2019-09.2020

Research Intern, Algorithm Engineer

Worked on visualization theory and intelligent methods for data
video generation and augmentation.

PUBLICATIONS

Asterisk* indicates equal contribution.

Junxiu Tang, Lijie Yao, Lu Ying, Romain Vuillemot, and Petra Isenberg. Diving Deep into Time: Temporal Arrangements for Embedded Visualization in Swimming Videos. In *IEEE Transactions on Visualization and Computer Graphics (TVCG) 2026*, DOI: 10.1109/TVCG.2026.3689361, OSF.

Junxiu Tang, Fumeng Yang, Jiang Wu, Yifang Wang, Jiayi Zhou, Xiwen Cai, Lingyun Yu, and Yingcai Wu. A Comparative Study on Fixed-order Event Sequence Visualizations: Gantt, Extended Gantt, and Stringline Charts. In *IEEE Transactions on Visualization and Computer Graphics (TVCG) 2024*, DOI: 10.1109/TVCG.2024.3358919, OSF.

Junxiu Tang, Yuhua Zhou, Tan Tang, Di Weng, Boyang Xie, Lingyun Yu, Huaqiang Zhang, and Yingcai Wu. A Visualization Approach for Monitoring Order Processing in E-Commerce Warehouse. In *IEEE Transactions on Visualization and Computer Graphics (TVCG) 2022 (Proceedings of IEEE VIS '21)*, DOI: 10.1109/TVCG.2021.3114878.

Junxiu Tang, Lingyun Yu, Tan Tang, Xinhuan Shu, Lu Ying, Yuhua Zhou, Peiran Ren, and Yingcai Wu. Narrative Transitions in Data Videos. Short Paper. In *Proceedings of IEEE VIS '20*, DOI: 10.1109/VIS47514.2020.00037, Project webpage.

Junxiu Tang, Rui Sheng, Yifang Wang, Xinhuan Shu, Xiaojiao Chen, Tan Tang, and Yingcai Wu. Posts with No Response: The Island of Loneliness. In *IEEE VIS Arts Program (VISAP2023@IEEE VIS)*, DOI: 10.1109/VISAP60414.2023.00007, OSF.

Junxiu Tang, Jiayi Zhou, Yifang Wang, Xinhuan Shu, Peiquan Xia, Xiaojiao Chen, Tan Tang, and Yingcai Wu. Loss of Sonnet 18. In *ACM SIGGRAPH Art Gallery (2024)*, DOI: 10.1145/3641523.3669939, OSF.

Junxiu Tang, Huimin Ren, Zikun Deng, Di Weng, Tan Tang, Lingyun Yu, Jie Bao, Yu Zheng, and Yingcai Wu. Towards Better Illegal Chemical Facility Detection with Hazardous Chemicals Transportation Trajectories. In *Journal of Visualization 2025 (Proceedings of ChinaVis '24)*, DOI: 10.1007/s12650-025-01055-8.

Tan Tang*, **Junxiu Tang***, Jiewen Lai, Lu Ying, Peiran Ren, Lingyun Yu, and Yingcai Wu. SmartShots: Enabling Automatic Generation of Videos with Data Visualizations Embedded. Extended Abstract. In *ACM International Conference on Multimedia (Proceedings of MM '20)*, DOI: 10.1145/3394171.3414356.

Tan Tang, **Junxiu Tang**, Jiewen Lai, Lu Ying, Yingcai Wu, Lingyun Yu, and Peiran Ren. SmartShots: An Optimization Approach for Generating Videos with Data Visualizations Embedded. In *ACM Transactions on Interactive Intelligent Systems (TiiS) 2022*, DOI: 10.1145/3484506.

Sadia Rubab, **Junxiu Tang**, and Yingcai Wu. Examining Interaction Techniques in Data Visualization Authoring Tools from the Perspective of Goals and Human Cognition: A Survey. In *Journal of Visualization 2021*, DOI: 10.1007/s12650-020-00705-3.

Tan Tang, **Junxiu Tang**, Jiayi Hong, Lingyun Yu, Peiran Ren, and Yingcai Wu. Design Guidelines for Augmenting Short-form Videos using Animated Data Visualizations. In *Journal of Visualization 2021 (Proceedings of ChinaVis '19)*, DOI: 10.1007/s12650-020-00644-z.

Jiayi Zhou, Renzhong Li, **Junxiu Tang**, Tan Tang, Haotian Li, Weiwei Cui, and Yingcai Wu. Understanding Nonlinear Collaboration between Human and AI Agents: A Co-design Framework for Creative Design. In *ACM CHI conference on Human Factors in Computing Systems (Proceedings of CHI '24)*, DOI: 10.1145/3613904.3642812.

Xinhuan Shu, Alexis Pister, **Junxiu Tang**, Fanny Chevalier, and Benjamin Bach. Does This Have a Particular Meaning? Interactive Pattern Explanation for Network Visualizations In *IEEE Transactions on Visualization and Computer Graphics (TVCG) 2024 (Proceedings of IEEE VIS '24)*, DOI: 10.1109/TVCG.2024.3456192.

Xinhuan Shu, Aoyu Wu, **Junxiu Tang**, Benjamin Bach, Yingcai Wu, and Huamin Qu. What Makes a Data-GIF Understandable? In *IEEE Transactions on Visualization and Computer Graphics (TVCG) 2021 (Proceedings of IEEE VIS '20)*, DOI: 10.1109/TVCG.2020.3030396, Project webpage.

Sadia Rubab, Lingyun Yu, **Junxiu Tang**, Yingcai Wu. Exploring Effective Relationships Between Visual-Audio Channels in Data Visualization. In *Journal of Visualization 2023*, DOI: 10.1007/s12650-023-00909-3.

Tan Tang, Kaijuan Yuan, **Junxiu Tang**, Yingcai Wu. Toward the Better Modeling and Visualization of Uncertainty for Streaming Data. In *Journal of Visualization 2018 (Proceedings of ChinaVis '18)*, DOI: 10.1007/s12650-018-0518-y.

Ran Chen, Xinhuan Shu, Jiahui Chen, Di Weng, **Junxiu Tang**, Siwei Fu, and Yingcai Wu. Nebula: A Coordinating Grammar of Graphics. In *IEEE Transactions on Visualization and Computer Graphics (TVCG) 2022*, DOI: 10.1109/TVCG.2021.3076222.

Yifang Wang, Yang Wang, Yifan Cao, Huamin Qu, **Junxiu Tang**, and Yingcai Wu. Explore Mindfulness without Deflection: A Data Art Based on the Books of Songs. Pictorial. In *Proceedings of IEEE VISAP '21*, DOI: 10.1109/VISAP52981.2021.00014, Project webpage.

ART EXHIBITION

Loss of Sonnet 18.

SIGGRAPH Art Gallery, Denver, Colorado, US

07.2024

ACADEMIC SERVICE

Posts with No Response: The Island of Loneliness.

China Vis Art Exhibition, Hong Kong, China	07.2024
Information+ Exhibition, Edinburgh, Scotland, UK	11.2023
IEEE VIS Art Exhibition, Melbourne, Victoria, Australia	10.2023

A Gear-like Storyline of Inception.

China Vis Art Exhibition, Shanghai, China	07.2018
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Co-organizer

What Do Visualization Art Projects Bring to the VIS Community?	2024
Panel @IEEE VIS 2024	

Program Committee

IEEE Visualization & Visual Analytics (VIS)	2026
IEEE VIS Arts Program	2025
IEEE Visualization & Visual Analytics (VIS) Short Papers, Posters	2025

Student Volunteer

IEEE Visualization & Visual Analytics (VIS)	2024
China Visualization and Visual Analytics Conference (ChinaVis)	2024
IEEE Eurographics / VGTC Symposium on Visualization (EuroVis)	2024
Information+ Conference (Info+)	2023

Conference Review

IEEE Visualization & Visual Analytics (VIS)	2022-2026
ACM Conference on Human Factors in Computing Systems (CHI)	2024-2026
ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW)	2025-2026
Creativity & Cognition	2026
ACM Symposium on User Interface Software and Technology (UIST)	2026
ACM Designing Interactive Systems (DIS)	2026
ACM Conference on Intelligent User Interfaces (ACM IUI)	2026
IEEE Pacific Visualization Symposium (PacificVis)	2024-2026
IEEE Eurographics / VGTC Symposium on Visualization (EuroVis)	2026
China Visualization and Visual Analytics Conference (ChinaVis)	2022-2025

Journal Review

IEEE Transactions on Visualization and Computer Graphics (TVCG)	2025-2026
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TEACHING

Zhejiang University

21191590: Cross Media Visualization	Winter 2022
21121280: Information Visualization	
Teaching Assistant	

Zhejiang University

International Visualization Summer School	Summer 2021
Teaching Assistant, Teaching Assistant Coordinator (TAC)	

Zhejiang University

International Visualization Summer School	Summer 2019-2020
Teaching Assistant	

TALK

Zhejiang University 21191590: Cross Media Visualization 21121280: Information Visualization Teaching Assistant, Teaching Assistant Coordinator (TAC)	Winter 2020
Intersections of Logic and Expression: Bridging Science and Art through Data. Boston, Massachusetts, US Conference talk @Data Art Symposium 2025	06.2025
A Comparative Study on Fixed-order Event Sequence Visualizations: Gantt, Extended Gantt, and Stringline Charts. Tampa, Florida, US (Virtual) Conference talk @VIS 2024	10.2024
From Concrete to Ambiguous and Reverse. Hangzhou, Zhejiang, China Invited talk @School of Art and Archaeology, Zhejiang University	09.2024
Loss of Sonnet 18. Denver, Colorado, US Conference artist talk @SIGGRAPH 2024	07.2024
Posts with No Response: The Island of Loneliness. Hong Kong, China Conference artist talk @ChinaVis 2024	07.2024
Towards Better Illegal Chemical Facility Detection with Hazardous Chemicals Transportation Trajectories. Hong Kong, China Conference talk @ChinaVis 2024	07.2024
Posts with No Response: The Island of Loneliness. Melbourne, Victoria, Australia Conference artist talk @IEEE VIS 2023	10.2023
Visualization x Art. Hangzhou, Zhejiang, China Invited talk @School of Design and Innovation, China Academy of Art	06.2023
A Visualization Approach for Monitoring Order Processing in E-Commerce Warehouse. New Orleans, Louisiana, US (Virtual) Conference talk @IEEE VIS 2021	10.2021
Narrative Transitions in Data Videos. Salt Lake City, Utah, US (Virtual) Conference talk @IEEE VIS 2020	10.2020

Design Guidelines for Augmenting Short-form Videos using Animated Data Visualizations.	
Chengdu, Sichuan, China	07.2019
Conference talk @China Vis 2019	

A Gear-like Storyline of Inception.	
Shanghai, China	07.2018
Conference talk @China Vis 2018	

Toward the Better Modeling and Visualization of Uncertainty for Streaming Data.	
Shanghai, China	07.2018
Conference talk @China Vis 2018	

MENTORSHIP

Zhejiang University	
Weichen Sun, Ph.D. student, co-mentored with Tan Tang	2024
Jiayi Zhou, undergraduate student, co-mentored with Yingcai Wu	2022-2024
Yuhua Zhou, Master student, co-mentored with Yingcai Wu	2020-2021
Boyang Xie, undergraduate student, co-mentored with Yingcai Wu	2020-2021
Lu Ying, undergraduate student, co-mentored with Yingcai Wu	2019-2020